ICRA: An Intelligent Clustering Routing Approach for UAV Ad Hoc Networks

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***Abstract*— As an important means of obtaining information of marine situation, the marine monitoring system relying on UAV has been paid more and more attention by all countries in the world, and the demand for tasks is growing continually. In UAV ad hoc networks, routing protocols with immutable routing policies that lack flexibility are generally incapable of maintaining effective performance due to the complicated and rapidly changing environmental situation and application requirements. In this paper, we propose an intelligent clustering routing approach (ICRA) for UANETs. The ICRA is composed of three components: the clustering module, the clustering strategy adjustment module and the routing module. In the clustering process, each node needs to calculate its utility. In order to maintain high topology stability and long network lifetime in different network states, the reinforcement learning based clustering strategy adjustment module needs continuous learning the benefits brought by adopting different strategies to calculate the nodes utility in a specific network state. With the learned knowledge, clustering strategy adjustment module could determine the optimal clustering strategy according to the current network state. In the routing phase, the proposed scheme can reduce the end-to-end delay and improve the packet delivery rate by introducing inter-cluster forwarding nodes to forward messages among different clusters. Extensive experiments have been conducted to verify ICRA's robustness and superiority over existing schemes. The results demonstrate that ICRA could achieve better performance than its state-of-the-art counterparts with regard to the clustering efficiency, topology stability, energy efficiency and quality of service.**

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I. Introduction

RECENT years have witnessed increasing development of unmanned aerial vehicle (UAV) networks in both civilian and military fields due to their characteristics of rapid networking, high autonomy, and strong anti-destruction [1]–[3]. Marine supervision tasks related to the national economy and people's livelihoods, such as national rights and interest protection, marine resource development and maritime safety protection, also need the support of UAV networks. Marine UAVs mainly provide services, including maintaining national marine safety and controlling marine pollution, through monitoring and investigation. Recently, maritime UAVs have also been used to perform cross-sea mail services, which can improve the convenience and timeliness of postal service in island areas that have imperfect transportation facilities.

In a UAV network, each node cooperates with other nodes to finish the predefined tasks. To manage a large number of UAV nodes and several static ground stations, self-organized UAV formation is a promising network structure, which is also called UAV ad hoc networks (UANETs) or flying ad hoc networks (FANETs) [4]. UANETs are a special form of mobile ad hoc networks (MANETs) and vehicular ad hoc networks (VANETs).

Due to the high movement speed and highly dynamic topology of UAV networks, routing protocols proposed for MANETs and VANETs cannot be directly applied to UANET environments. For data transmission between UAV nodes, clustering-based routing protocols are used increasingly widely as the number of UAVs increases. In these protocols, the UANET has a hierarchical structure in which a node can be a cluster head (CH) or a cluster member (CM). CMs need to fly according to the scheduled route and complete the tasks (e.g., monitoring, military strike and so on) assigned to them. In addition to completing the same functions as the CMs, the CHs also need to manage the CMs within the cluster and act as a communication relay for intra-cluster and inter-cluster communication. Only the elected CH will be responsible for communication with the ground station. Due to the dynamic nature of the mobile nodes in UANETs, their association and dissociation to and from clusters perturb the stability of the network topology, and thus, reconfiguration of CHs is unavoidable. This issue is important because frequent topology

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changes will affect the reliability of the routing, resulting in routing inefficiency, thus negatively affecting the effect of upper layer services.

Generally, these available clustering-based routing protocols for UANETs mainly include two phases, namely, clustering and routing. The clustering phase is mainly in charge of the network topology construction, and the routing phase addresses the end-to-end communication link establishment between nodes. During the clustering process, the CH nodes should be first determined. Beforehand, some factors are chosen to measure the utility of each node. Then, the node that has the highest utility within a certain range will be chosen as a CH. The most commonly used utility factors are the node residual energy, node centrality, node degree, and so on. Several studies adopt only one factor (e.g., the geographical position or logical position of each node [5]–[7], the connectivity ability of each node [8], [9]) as the utility factor to elect CH nodes. Some studies consider multiple factors in the cluster formation phase [10]–[12]. In these schemes, each factor is assigned a certain weight, and the weighted sum of all factors of each node is treated as its utility. The node with the highest utility within a certain range will be selected as the CH node in this range. For the value of the utility factor weights, most studies point out that the sum of all of the weights should be 1. Some protocols assign the same value for the weight of each factor, while some protocols specify different weights for each factor. What these schemes have in common is that the weights of the different factors are fixed during the whole network lifetime.

In some simple task scenarios, the UAV network is composed of the same type of UAV nodes, and each node performs the same task. In some complex task scenarios, the UAV network is composed of multiple types of UAVs with different tasks. Each node is deployed with different resources and has different resource consumption rates and flying routes. In addition, in some scenarios (such as wireless coverage), UAVs will be deployed statically; in other words, UAVs will be evenly distributed and suspended in a fixed position to complete the task in a given area after arriving at the workplace along a given route. However, in most complex scenarios, UAV networks are deployed dynamically. All of the nodes will move in a given area according to a certain mode, with high speed and different mobility. The essential difference for the abovementioned scenarios in terms of the routing protocol is the value distribution of the utility factors used by all nodes in CH elections. For example, if a network with all nodes has the same initial energy and energy consumption rate, the probability distribution of the residual energy of all of the nodes follows $\forall n \in NU, P(E_n^t = E_{n'}^t | n \neq n', n' \in NU) = 1$. Here, n and n' represent any node in the network, NU means the set of nodes in the network, and E_n^t is the residual energy of node n at time t. If a few nodes in a network have much initial energy, while other nodes have limited energy, and their energy consumption rates are different, then the probability distribution of the residual energy of all nodes follows $\forall n \in NU, P(E_n^t = E_{n'}^t | n \neq n', n' \in NU) < 1$. The meaning of symbols here is the same as that in the above equation.

When the utility of each node is measured during the CH election, it is clear that the importance of each factor should vary with the distribution of the value of this factor in all of the nodes in the network. When the residual energy of all nodes is the same in the whole life cycle of the network, those factors with wider value distributions deserve more attention. However, if the residual energy of each node is different from the others, the more dispersed the value of all nodes' residual energy is, the more sensitive the value of a node's utility in acting as a CH will be to its residual energy. In most of the existing work, the weight of each utility factor is determined in advance when calculating the utility value. In the network lifetime, regardless of how the value distribution of each factor changes, the weight will not change. As a result, the static CH election scheme cannot be adjusted adaptively in the dynamic network state scenario, resulting in failure to obtain the optimal or approximate optimal performance, such as the stability of the network topology and the network lifetime.

To solve the problem shown above, we design a CH node election method based on a reinforcement learning model to maintain a stable network topology in a dynamic environment. In this method, the reinforcement learning model is used to learn the optimal or approximate optimal node utility measurement strategy (i.e., the weight value of each factor in the utility calculation) under different network states in such a way that UAV nodes can cluster according to the real-time state of the network by adopting the strategy that makes the network topology more stable. After a stable network topology is obtained, we propose a routing method for nodes in the same cluster and nodes in different clusters, which makes the communication between nodes have a low end-to-end delay and a high packet delivery rate. The proposed approach provides a stable and reliable transmission basis for the business of UANETs.

In conclusion, the contributions of this paper are threefold.

- We propose an intelligent routing framework for hierarchical UANETs, which can maintain stable network topology and a high package delivery ratio with varying network states.
- We exploit a highly efficient CH election method that uses reinforcement learning, with which the nodes in the network is able to constantly obtain the optimal CH node election strategy during the network lifetime.
- We propose an intra-cluster and inter-cluster routing method, with which the network can have a low end-to-end package delivery delay and a high package delivery ratio.

The remainder of this paper is organized as follows. Section II shows the related work of this paper. Section III introduces the proposed method in detail. In section IV, we present the experiments that we performed to verify the efficiency of our approach. Finally, Section V concludes this paper.

II. Related Work

In recent years, UAV ad hoc networks have been a focus area of the academic and industrial research community. They have been used in many military and civilian

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applications [13], [14]. In UAV ad hoc networks, routing is a challenging task. Although a UANET is a subdomain of ad hoc networks, its unique characteristics do not allow one to use MANET and VANET routing schemes directly [15]–[17]. A plethora of research work has been conducted recently to investigate and tackle robust and reliable routing approaches for UANETs [18]. In this section, we review and discuss the existing routing techniques for UANETs.

The topological structure of UANETs can be divided into two categories, namely, nonhierarchical and hierarchical structures. To ensure the scalability of the network, a hierarchical structure is generally adopted in large- and medium-sized UANETs. Farmani *et al.* proposed a DBSCAN (density-based spatial clustering of applications with noise)-based clustering algorithm for UAV networks [5]. Using the estimated positions of UAVs and specifying the DBSCAN input parameters, UAVs can dynamically be grouped into clusters. Khelifi *et al.* proposed a localization-based mechanism for data routing in UAV networks [6]. The proposed localization algorithm first localizes the UAVs and then elects the CH to minimize the energy consumption of the whole network and consequently increase its lifetime. Lin *et al.* proposed a logical position-based shortest path routing protocol for UANETs, in which the logical grid distance is used to replace the original Euclidean distance in the routing process [7]. It can be seen that these protocols mainly consider a certain metric of nodes to construct the network topology and routing path. In addition, the most commonly considered clustering factor is the position of the nodes.

Recently, an increasing number of researchers have considered multiple factors in selecting the CH nodes in the clustering process. Qi *et al.* proposed a clustering architecture for a flying ad hoc sensor network [8]. In their work, each cluster of the network executes a centralized traffic-differentiated routing, which considers several factors, including the link availability and the nodes' forwarding ability. Different weights are assigned to these factors based on the sensitivity of the network topology stability. Shi and Luo proposed a cluster-based location-aided dynamic source routing (CBLADSR) scheme [10], which is composed of clustering, routing discovery and maintenance processes. In the clustering process, multiple factors, including the value of the connectivity degree, relative speed, residual energy, and equipment-dependent tactical of the nodes, are considered in selecting the CH nodes. The node that satisfies the residual energy threshold and has the highest weighted sum of all factor values will become a CH. The weight of each factor is determined by application scenarios, and the sum of all factor weights should be equal to 1. Zhang and Zhang proposed a mobility prediction clustering algorithm (MPCA). In the clustering process, the connectivity of the nodes and the probability of their neighborhood set remaining in the next stage are considered [11]. Aadil *et al.* proposed an energy aware cluster-based routing protocol for FANETs [19]. In their work, the K-Means density clustering algorithm was used for CH selection. A fitness value is used to measure whether a node can be selected as a CH, which is calculated based on three factors, namely, the average distance of a node to its neighboring nodes, the number of its neighbors and its residual energy. The authors use the reciprocal of a factor's deviation from the mean value of all factors as the weight of this factor. Yu *et al.* proposed a clustering-based communication backbone for large-scale UAV networks [20]. The core idea of this work is to build a cluster of UAVs. The CH of a cluster is elected by calculating the average distance between a node and its neighbors. The schemes mentioned above take various factors into account for CH selection in the clustering process, while most of them do not give a concrete method to determine the weight of each factor. For the schemes that give the method for calculating the weight of each factor, the authors all default in that the weight of each factor will remain static throughout the network life cycle.

Recently, motivated by its successful applications to many practical tasks such as image recognition, both industry and the research community have advocated for applications of machine learning in wireless communication, including the MAC layer, network layer and application layer [21]–[28]. Specifically, some researchers have applied reinforcement learning theory to routing protocols for various wireless networks. Reinforcement learning model is used to deal with the continuous strategy optimization. It considers tasks in which the agent interacts with an environment through a sequence of observations, actions, and rewards. The goal of the agent is to select actions in a fashion that maximizes the cumulative future reward. In [29], the authors proposed a Q-learning-based geographic routing scheme to improve the network performance for unmanned robotic networks [30]. Saleem *et al.* also used reinforcement learning technology to address the clustering and routing problem for cognitive radio networks [31]. However, to the best of our knowledge, there is no cluster-based routing protocol for UANETs that uses reinforcement learning to optimize the network topology and routing performance.

Considering all of the above work, we aim to adopt reinforcement learning in the routing scheme for hierarchical UANETs to adjust the clustering strategy wisely as the network states change. Compared with available work about clustering routing protocols for UANETs, the proposed clustering strategy is able to enhance the stability of the network topology and extend the network lifetime in different scenarios. The resulting stable network topology can support our proposed routing scheme to achieve a high packet delivery rate and low end-to-end delay.

III. Proposed Scheme

In this section, we will introduce the system framework and the proposed scheme in detail.

A. Network Model and Design Goal

The UAV ad hoc networks considered in this work generally refer to a network that consists of a ground station and several UAV nodes, which can be indicated by $< GS, NU >$, where GS is the ground station, and NU is the set of UAV nodes. With a certain clustering algorithm, nodes within NU can be divided into two types of entities, namely, cluster

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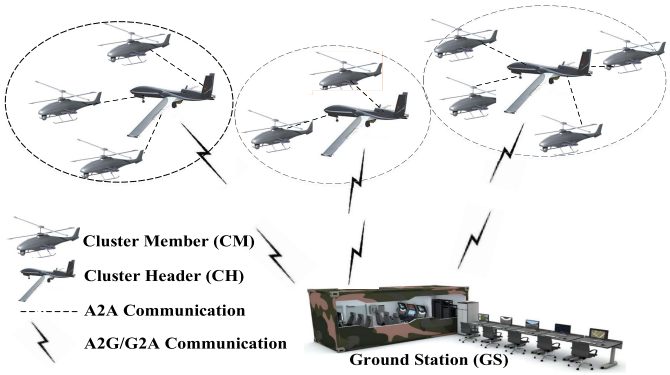

Fig. 1. The network model of the proposed scheme.

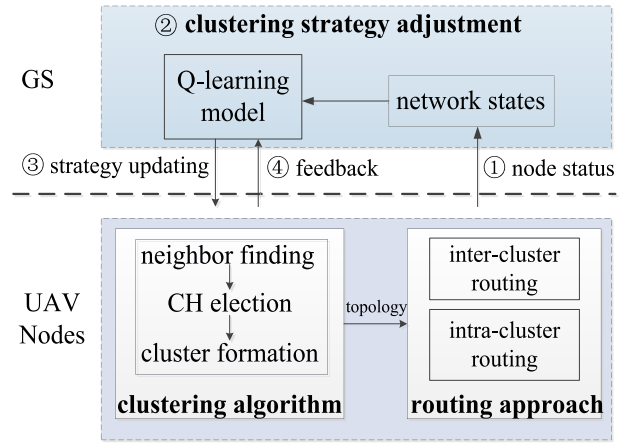

Fig. 2. The proposed framework.

heads (CHs) and cluster members (CMs). In the following, we will collectively refer to CH nodes and CM nodes as nodes or entities of the network. The network model considered in our work is shown in Fig. 1. In general, there are two types of communication in the considered network, namely, communication between UAVs and the ground (including air-to-ground (A2G) and ground-to-air (G2A)) and communication between UAVs (air-to-air, A2A). Each UAV node has its own flight route and tasks to be performed. We assume that there is no resource limitation in the ground station. In addition, each UAV node has different capabilities and resources, and has its own flight route and mission to perform. If a UAV node is within the communication range of the GS, they can communicate with each other. It is assumed that all nodes have a GPS with a network time protocol system. The GPS provides the location and mobility of a node to calculate the link expiration time between two nodes. The responsibility of each type of entity in a UANET is shown below.

GS: As the command center of the whole UAV system, the control content of the ground station includes each UAV node's flight process, flight path, mission function of the payload, normal operation of the communication link, etc.

CMs: Each UAV node flies in a specific flight mode and performs its predefined tasks. In addition, UAV nodes cooperate with each other to make up the adaptive network.

CHs: CHs, in addition to completing the functions of CMs, are also responsible for the management of nodes within their clusters.

In our scheme, all UAV nodes should first be clustered into several groups using the clustering algorithm to form the network topology. Based on the network topology and the routing approach, the communication route between any nodes can be determined. The clustering process can be expressed by Eq. 1, where F_{clt} represents the clustering algorithm. The first parameter of F_{clt} represents the clustering strategy at time slot t. The output G_t is the network topology at time slot t, which can be denoted as a set of tuples shown in Eq. 2. In Eq. 2, $< CH_i^t, CM_i^t >$ is the ith cluster of the network at time slot t, where CH_i^t and CM_i^t are the CH and the set of CMs of this cluster. It is obvious that the union of CH set CH_t (shown in Eq. 3) and cluster members set CM_t (shown in Eq. 4) is equal to NU (shown in Eq. 5).

$$G_t = F_{clt}(S_t, NU) \tag{1}$$

$$G_t = \{< CH_i^t, CM_i^t > | i \in [1, n_t]\} \tag{2}$$

$$CH_t = \{CH_i^t | i \in [1, n_t]\} \tag{3}$$

$$CM_t = \{CM_i^t | i \in [1, n_t]\} \tag{4}$$

$$NU = CH_t \cup CM_t \tag{5}$$

For the design goal of this work, the first goal is to propose a self-adapting clustering framework which can maintain a stable network topology and long network lifetime in different network states. The second goal is to propose an efficient routing approach to maintain a low packet loss rate and high packet delivery ratio among the nodes to make the network communications more reliable.

B. System Framework

Fig. 2 shows the proposed ICRA framework. It can be seen that the whole framework is composed of three modules, namely, the clustering module, clustering strategy adjustment module and routing module.

The clustering module is in charge of forming the entities within the UAV network as clusters based on a clustering algorithm. The first phase of the clustering algorithm is neighbor finding. Each entity broadcasts its information to and receives messages from its neighbors to obtain information about its neighbors. Then, according to a certain clustering strategy, each entity calculates utility values of itself and its neighbors. Then, CH nodes will be elected based on these utility values. Finally, other nodes will determine which CH they belong to. Thus, the hierarchical topology of the UAV network is formed.

Over time, the status (including the speed, remaining energy, etc.) of each entity will change accordingly, and thus, the state distribution of the nodes in the network will also change continuously. This process will cause the criteria for measuring a node's utility should also be adjusted. In such a way, the network topology formed based on the utility of all nodes can be more stable and the network lifetime can be longer. In the proposed framework, the module used to complete the

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node utility measurement criteria adjustment task mentioned above is the clustering strategy adjustment module. The core of the clustering strategy adjustment module is a reinforcement learning model, which constantly learns the optimal or approximate optimal clustering strategy under different network states based on feedback from the previous routing result. Furthermore, it periodically evaluates the network states based on the received nodes' status. According to the current network status, the optimal or near optimal strategy can be selected and then sent to the network entities. Based on the latest strategy, entities of the network are able to construct the most suitable network topology for the current network state.

Given the network topology, we need a routing approach to determine the communication route between any sender and receiver. Based on whether the message sending node and the receiving node belong to the same cluster, we design the routing method between nodes within the same cluster (intra-cluster) and the routing method between nodes that belong to different clusters (inter-cluster).

At the beginning, the network is assigned a default clustering strategy. Then, the network will operate according to the following processes, which involves all the modules of the proposed framework.

- Step 1. Entities within the UAV network construct the network topology based on a given clustering strategy. Based on the network topology and routing algorithm, a network node is able to determine the communicate path with other nodes.
- Step 2. Each entity sends its status information and the feedback of the clustering result at regular intervals to the clustering strategy adjustment module deployed in the GS.
- Step 3. The feedback information will be seen as the reward of the reinforcement learning model, and the clustering strategy adjustment module will update its learning model correspondingly. Meanwhile, the clustering strategy adjustment module updates the network states using the received entity status. Based on the latest network state, the clustering strategy adjustment module determines the optimal clustering strategy for the network, and then forwards the clustering strategy to the clustering module of UAV nodes.

These three steps are conducted repeatedly, maintaining the topology construction and communication route establishment of the whole network. In the following sections, we will describe the concrete algorithms included in these modules in detail. For ease of discussion, we list the symbols used in the following sections and their meanings in TABLE I.

TABLE I
MEANING OF SYMBOLS USED IN THE FOLLOWING SECTIONS

Symbol	Meaning
N	The number of nodes in the UAV network.
ID_i	Identity of the ith node in the network.
W_table_i	Node ID_i's utility message list.
CH_table_i	Node ID_i's message list to store statement message from CH nodes.
N_table_i	Node ID_i's neighbor list to store its neighbors' information.
M_table_i	Cluster head ID_i stores information of its member nodes in this list.
$P_i = \{La_i, Lo_i\}$	Node ID_i's geographical position, where La_i and Lo_i represent its latitude and longitude respectively.
$\vec{S}_i = \{v, \theta\}$	Node ID_i's velocity, where v and θ mean the current speed and flight angle, respectively.
R_i	Role of node ID_i. If $R_i = 0$, then ID_i is a CH; if $R_i = x$, then the CH of the cluster to which ID_i belongs is ID_x.
W_i	Current utility value of node ID_i.
s_i^k	kth utility factor of ID_i. $k = 1, 2, 3, 4$
w_k	The weight for factor s_i^k.
$LHTP^{i,j}$	Link holding time between ID_i and ID_j.
E_{hf}	Energy consumption rate for CH and forwarding nodes.
E_n	Energy consumption rate for common cluster member nodes.

C. Clustering Algorithm

The clustering process includes five steps. In this section, we first show the categories of messages that will be transmitted between two nodes ID_i and ID_j ($i \neq j$) during the clustering process. Then, we introduce the details of each step in the following subsections.

- **Hello message**. Node ID_i broadcasts its basic information to its neighbors by this message, which can be formalized as $\boldsymbol{Hello_i = ID_i || P_i || \vec{S}_i || R_i || timestamp}$.
- **Utility message**. $\boldsymbol{M_i = ID_i || W_i || timestamp}$. Node ID_i sends its utility value W_i by this message.
- **CH declaration message**. $\boldsymbol{CH_i = ID_i || W_i || CH || timestamp}$. Node ID_i broadcasts this message to declare that its role is a CH.
- **Join cluster request message**. $\boldsymbol{join(i, j) = ID_i || ID_j || REQ_JOIN || timestamp}$. This message is sent by node ID_i, which represents node ID_i requests to join the cluster in which node ID_j is the CH.
- **Join cluster response message**. $\boldsymbol{res_{join}(j, i) = ID_j || ID_i || RES_JOIN || timestamp}$. This message is sent by node ID_j, which means that node ID_j allows node ID_i to join its cluster.

1) Step1 Network Initialization: At the beginning, node ID_i ($i \in [1, N]$) in the UAV network broadcasts $\boldsymbol{Hello_i}$ to its neighbors. For all received hello messages $\{Hello_j | j \neq i, j \in [1, N]\}$, ID_i extracts the information included and stores its neighbors' information to list N_table_i.

2) Step2 Cluster Head Election: In order to select the cluster head which is conducive to more stable network topology and longer network lifetime, we need to calculate the utility value of each node, and select the node with the largest utility value within a certain range as the cluster head. In this work, we take four factors into account to calculate the utility of a node, namely, the residual energy of the node, the degree centrality of the node, the velocity similarity between the node and its neighboring nodes, and the link holding time between the node and its neighboring nodes. Each node ID_i should follow the steps shown below to complete the CH election.

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Step 2.1. ID_i calculates the value of its four utility factors $s_i^k (k = 1, 2, 3, 4)$, using the information of itself and its neighbors. The details of the calculation method for each utility factor will be introduced in the following section.

Step 2.2. ID_i obtains the value of its utility W_i by Eq. 6. In Eq. 6, W_i is the weighted sum of the four utility factors $s_i^k, (k = 1, 2, 3, 4)$. For the weights of the four utility factors, we have $w_k \in [0, 1]$ and $\sum_{k=1}^{4} w_k = 1$. The method of determining the weight w of each utility factor is in the charge of the strategy adjustment module, which will be introduced in the following content.

$$W_i = \sum_{k=1}^{4} w_k \times s_i^k \tag{6}$$

Step 2.3. ID_i broadcasts utility message M_i to and receives utility messages from other nodes. ID_i stores the received utility messages M_j $(j \neq i)$ in list W_table_i. If there is no utility value in W_table_i larger than W_i, ID_i will be a CH, and its role indicator R_i will be set to 0. Then, ID_i broadcasts **CH declaration message** CH_i. Otherwise, go to step 3.

3) Step 3 Cluster Formation: After selecting the CH node, we need to determine the cluster members in each cluster. The method for cluster formation will determine the cluster to which each CM belongs. The details of the method are shown in Algorithm 1.

Algorithm 1 Cluster Formation Algorithm

Require: W_i, CH_table_i, $(i \in [1, N])$
Ensure: R_i // role of ID_i
create integer $k = 0$, $q = 0$;
// constantly receiving **CH declaration messages** from other nodes
while ID_i receives message CH_j $(j \neq i)$ **do**
  insert CH_j into list CH_table_i;
end while
$q = sizeof(CH_table_i)$;
for $k = 0 \sim q - 1$ **do**
  W_k is the second field of the kth element in CH_table_i;
  if $W_i > W_k$ && $LHTP^{i,k} > \sigma$ **then**
    // σ is a predefined threshold.
    ID_i sends $join(i, k)$ to ID_k;
  end if
  if ID_i receives $res_{join}(k, i)$ from ID_k **then**
    $R_i = k$; //ID_i's CH is ID_k.
    break;
  end if
end for
if $k == q$ **then**
  $R_i = 0$; //ID_i does not join any cluster.
  ID_i broadcasts **CH declaration message** CH_i;
end if
return R_i;

4) Step 4 Network Topology Establishment: In this step, each node in the network will determine its one-hop communication node.

For a CM node, from the Hello messages it receives, it can obtain the identity of its neighborsćluster heads. If a node finds that its neighbor ID_j does not belong to the same cluster as itself, it will apply to its CH to become an inter-cluster forwarding node. In this way, the UAV network is composed of three types of nodes in total, namely, the cluster heads, inter-cluster forwarding nodes, and common cluster member nodes. Here, inter-cluster forwarding nodes and common cluster member nodes are collectively referred to as CM nodes.

Cluster head node and inter cluster forwarding node are forwarding nodes in the network, which constitute the backbone of the network. For a CH node, its one-hop communication entities include other CHs within its communication range and the CMs within its cluster. The one-hop neighbors of an inter-cluster forwarding node include its CH and its neighbors that not belong to the same cluster as itself. For the forwarding nodes, we are able to not only send messages directly to their one-hop entities, but also forward messages for them. While, a common cluster member nodes can only communicate directly with its CH.

5) Step 5 Cluster Maintaining: Due to the dynamic topology (nodes joining and leaving the network, flight path changes, and so on), all entities within the UANET must participate in the cluster maintaining task cooperatively. The aim of cluster maintenance is to dynamically adjust the role of each node based on regular information exchange, to guarantee the connectivity of the network topology. In the following, we present the cluster maintaining method under different situations.

1 **Node joining**. If a node ID_i joins the UANET, it will broadcast **join cluster request message** $join(i, 0)$, where the second parameter 0 represents that there is no target CH. The CH node that responds to the request first will become the CH of ID_i.
2 **Node leaving**. Each CM node ID_i within the UANET must regularly check whether the communication link with its CH is valid. If it cannot communicate with its CH, it will exit the cluster and enter Step 3 (described in Section III-C.3) to reselect its CH.
3 **Cluster member maintaining**. For any CH node ID_i, it should regularly check whether the communication link with the nodes stored in its member list M_table_i is valid. Nodes with invalid communication links will be deleted from M_table_i, and these nodes will no longer belong to this cluster.
4 **Communication backbone maintaining**. All nodes within the UANET send and receive **hello message** regularly. Based on the obtained information and the conditions that the intercluster forwarding node must meet, each node can dynamically decide whether to act as an intercluster forwarding node.
5 **Re-clustering**. If nodes within the UANET receive the clustering strategy update message, they will enter Step 1 (described in Section III-C.1) to restructure the clustering according to the latest clustering strategy.

6) Utility Calculation Method: As mentioned in Section III-C.2, in the CH election process, each node

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must calculate its utility to determine whether to declare it as a CH. In this section, we present the method for calculating the utility for a node.

Eq. 6 shows that the utility of ID_i is the weighted sum of four utility factors (s_i^k, $k = 1, 2, 3, 4$). In the following, we give the meaning of these four utility factors and their calculation method.

- $\boldsymbol{s_i^1}$. The first utility factor is used to measure the residual energy of a node. It can be calculated by Eq. 7, where E_i^t means the residual energy of ID_i at time slot t, and E_i^0 is the initial energy of ID_i.

$$s_i^1 = \frac{E_i^t}{E_i^0} \tag{7}$$

- $\boldsymbol{s_i^2}$. The second utility factor is used to measure the degree centrality of a node in the network. Eq. 8 shows how to calculate s_i^2, where D_i^t represents the degree of ID_i at time slot t, and N is the number of nodes in the network.

$$s_i^2 = \frac{D_i^t}{N-1} \tag{8}$$

- $\boldsymbol{s_i^3}$. The third utility factor is the velocity similarity of a node with its neighbors, which can be obtained by Eq. 9, 10, 11 and 12.

  We assume that at time slot t, the velocity of ID_i and ID_j $(i \neq j)$ is $\overrightarrow{S_i^t} = \{v_i^t, \theta_i^t\}$ and $\overrightarrow{S_j^t} = \{v_j^t, \theta_j^t\}$ separately. In Eq. 9, $d_{i,j}^t$ is the Euclidean distance of ID_i and ID_j's velocity at time slot t.

  In Eq. 10, $V_{i,j}^t$ represents the velocity similarity between ID_i and ID_j at time slot t. The larger the value of $d_{i,j}^t$ is, the smaller the velocity similarity between nodes ID_i and ID_j.

  $\bar{V}_{s(i)}^t$ in Eq. 11 is the average velocity similarity between ID_i and its neighboring nodes at time slot t. In addition, symbol n in Eq. 11 and 12 means the number of ID_i's neighbor nodes.

$$d_{i,j}^t = \sqrt{(x_i^t - x_j^t)^2 + (\theta_i^t - \theta_j^t)^2} \tag{9}$$

$$V_{i,j}^t = \frac{1}{1 + d_{i,j}^t} \tag{10}$$

$$\bar{V}_{s(i)}^t = \frac{1}{n} \cdot \sum_{j \in N_table_i} V_{i,j}^t \tag{11}$$

$$s_i^3 = \frac{1}{n} \cdot \sum_{j \in N_table_i} (V_{i,j}^t - \bar{V}_{s(i)}^t)^2 \tag{12}$$

- $\boldsymbol{s_i^4}$. The last factor s_i^4 is used to measure the link holding time between ID_i and its neighboring nodes. Eq. 13 and 14 show the calculation method of s_i^4, where $LHTP_{i,j}^t$ means the link holding time between ID_i and ID_j at time slot t.

$$LHTP_{i,j}^t = \frac{-(ab + cd) + \sqrt{(a^2 + c^2)R^2 - (ad - bc)^2}}{a^2 + c^2} \tag{13}$$

$$s_i^4 = \frac{1}{n} \sum_{j \in N_table_i} LHTP_{i,j}^t \tag{14}$$

Eq. 15, 16, 17 and 18 give the calculation method for variables a, b, c and d in Eq. 13. Here, we use (X_i, Y_i) and (X_j, Y_j) to represent the positions of ID_i and ID_j, and $< \nu_i, \theta_i >$ and $< \nu_j, \theta_j >$ indicate their velocities.

$$a = \nu_i \cos\theta_i - \nu_j \cos\theta_j \tag{15}$$

$$b = X_i - X_j \tag{16}$$

$$c = \nu_i \sin_i - \nu_j \sin_j \tag{17}$$

$$d = Y_i - Y_j \tag{18}$$

D. Reinforcement Learning-Based Clustering Strategy Adjustment Approach

To ensure that the UAV ad hoc network maintains high efficient routing in dynamic network scenarios, we use a reinforcement learning model based on the Q-learning algorithm to adjust the optimal clustering strategy according to the current network state. The Q-learning algorithm was proposed by DeepMind [32], [33] and used to address continuous strategy optimization. It considers tasks in which the agent interacts with an environment through a sequence of observations, actions and rewards. The goal of the agent is to select actions in a fashion that maximizes the cumulative future reward. The GS acts as the clustering strategy adjustment server to run the reinforcement learning model and broadcasts the latest optimal clustering strategy to all nodes regularly. UAV nodes within the network will restructure according to the received clustering strategy. Then, feedback information will be sent back to the GS based on the clustering results, with which GS will update its reinforcement learning model (Q-table).

In the following, we will first introduce the definition of the status of a UAV node. Then, we will give the action space, state space, incentive function and the strategy adjustment method of the clustering strategy adjustment model.

Status of a UAV Node: We use a quad that consists of these four utility factors mentioned in the previous section to represent the status of a node. It can be formalized as Eq. 19, where the states of ID_i is denoted as sn_i.

$$sn_i = < s_i^1, s_i^2, s_i^3, s_i^4 > \tag{19}$$

Action Space: The action space of the reinforcement learning model can be formalized as a quad $AS = < w_1, w_2, w_3, w_4 >$, where w_i $(i \in [1, 4])$ represents the weight of the ith utility factor. The value domain of each element of AS is $w_i = \{0.05 * p | p \in [0, 20]\} (i \in [1, 4])$. In addition, the value of w_i should satisfy $w_1 + w_2 + w_3 + w_4 = 1$.

State Space: At time slot t, the state of the whole network is denoted as S^t, which can be formulated as a vector $S^t = < s_1^t, s_2^t, s_3^t, s_4^t >$. Here, each of the elements s_i^t in S^t can be obtained by Eq. 20, where H_i means the entropy of the ith element of all nodes' status vector. From Eq. 20, it can be seen that the state space of the reinforcement learning model is $SS = \{< 0.1 \cdot i, 0.1 \cdot i, 0.1 \cdot i, 0.1 \cdot i > | i \in [0, 10]\}$.

For the ith $(i \in [1, 4])$ utility factor (the ith element of a node's status vector shown in Eq. 19), we assume that the set of its values for all nodes in the network is $U_i^t = \{u_i^1, u_i^2, \cdots, u_i^N\}$, and the probability of $0.1 * x (x \in$

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[0, 10]) appearing in U_i^t is denoted as P_i^x. On this basis, Eq. 21 shows the calculation method for H_i.

$$s_i^t = \lfloor H_i \ mod \ 0.1 \rfloor \tag{20}$$

$$H_i = -\sum_{x=0}^{10} (P_i^x \cdot log P_i^x), \quad i \in [1, 4] \tag{21}$$

Reward Function: In our clustering strategy adjustment model, the GS continuously receives feedback on the effect of the clustering results sent by all nodes in the network. The feedback will be seen as the reward to the strategy used in the latest clustering process.

Eq. 22 gives the function to calculate the reward, which is denoted as $R(r)$. The parameter r is used to measure the stability of the clustering structure (denoted as R_c) and the energy change of the nodes (denoted as E_c) during a period of time T.

$$R(r) = \begin{cases} 1 & r = 1 \\ -1 & r = -1 \\ \frac{1-e^{-3r}}{1+e^{3r}} & otherwise \end{cases} \tag{22}$$

Eq. 23 gives the method for calculating r, where λ represents the weight of the cluster structure stability. Here, we use the number of times a node changes roles over a period of time t, which is denoted as $count_role$, to measure the structural stability of the cluster to which it belongs. If the number of times a node changes roles is less than a predefined threshold φ, the value of R_c will be 1; otherwise, it will be set to -1, which is shown in Eq. 24. Eq. 25 and 26 show the calculation method for E_c, where ΔE means the change rate of the node energy in a period of time T. It is obvious that the range of R_c and E_c is $[-1, 1]$.

$$r = \lambda \times R_c + (1 - \lambda) \times E_c \tag{23}$$

$$R_c = \begin{cases} 1 & count_role < \varphi \\ -1 & otherwise \end{cases} \tag{24}$$

$$E_c = 1 - 2\Delta E \tag{25}$$

$$\Delta E = \frac{E_i^{t-T} - E_i^t}{E_i^0} \tag{26}$$

After giving the state space, action space and reward function used in our clustering strategy adjustment model, Algorithm 2 shows the learning process in detail. In Algorithm 2, $Q[s, a]$ stores the approximate optimal total revenue value if the network takes action a at state s. $R[s, a]$ records the reward when UAV nodes take clustering strategy a at network state s. It is the average value of feedback from all UAV nodes ($R(r)$ in Eq. 22) for taking action a at state s during period T. Here, taking action a means that the UAV nodes use the value of a as weights of the four utility factors during the clustering process. Lines 1 $\sim$ 8 represents the initial phase. We assume at the beginning that each utility factor has the same importance when calculating the utility of a node. Therefore, we set $Q[s, < 0.25, 0.25, 0.25, 0.25 >] = 1$ so that the initial optimal clustering strategy is $< 0.25, 0.25, 0.25, 0.25 >$. Lines 9 $\sim$ 18 show the learning phase that follows the Q-learning algorithm, which makes the value of elements in the Q-table vary with the feedback from the nodes. In line 16, parameter α is the learning rate, which has a value in the range [0, 1]. The larger α is, the less effect that the historical learning result will have on the learning process, and vice versa.

Notably, in line 16, the update function for the Q-table is $Q[s, a] = \alpha(R[s, a] + \gamma \times max_{\alpha' \in AS} Q[s_{next}, \alpha']) + (1 - \alpha) \times Q[s, a]$, where s_{next} is the next state of the network after taking an action a in state s, and γ is the discount factor. The larger the value of γ is, the more the future reward is considered in the learning process. However, in our model, we know that the state of the network is determined by all nodes' statuses, and thus, given (s, a), s_{next} is not deterministic. Therefore, in Algorithm 2, we set $\gamma = 0$, which means that we consider only the historical learning result and the current reward to update the Q-table ($Q[s, a]$).

Algorithm 2 Clustering Strategy Learning Algorithm

Require: Feedback (reward) for an action on a certain state.
State space set SS.
Action space set AS.
Ensure: The approximated optimal action-value function $S(s, a)$.

1: create two tables, Q-table $Q[s, a]$ and R-table $R[s, a]$, $s \in SS \cap a \in AS$;
2: **for** $s \in SS$ **do**
3: **for** $a \in AS$ **do**
4: $Q[s, a] = 0$; $R[s, a] = 0$;
5: **end for**
6: $Q[s, < 0.25, 0.25, 0.25, 0.25 >] = 1$;
7: **end for**
8: $rd = 0$; $n = 0$;
9: **for** the whole process of learning **do**
10: **for** feedback R for action a at state s received during period T **do**
11: $rd += R$;
12: $n++$;
13: **end for**
14: $rd = \frac{rd}{n}$;
15: $R[s, a] = rd$;
16: $Q[s, a] = \alpha(R[s, a] + \gamma \times max_{\alpha' \in AS} Q[s_{next}, \alpha']) + (1 - \alpha) \times Q[s, a]$;
17: $T = T + 1$;
18: **end for**
19: **for** $s \in ss$ and $s \in as$ **do**
20: $S(s, a) = Q[s, a]$;
21: **end for**

At the end of each time slot, the GS will calculate the network state s and then select the optimal strategy (action) by Eq. 27. If there is more than one action that satisfies Eq. 27, the GS should choose one of them randomly with the same probability.

$$a = \{a' | Q[s, a'] = max\{Q[s, a] | a \in AS\}\} \tag{27}$$

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E. Routing Approach

With the established network topology, nodes within the UANET will communicate with each other based on a certain routing protocol.

To conduct the routing approach, each node within the network should store its information, including its identity ID, position, velocity, link holding time and velocity similarity with its neighbors, as well as its role in the topology. In the following, we will introduce the routing approach in detail.

Step1. When node ID_i receives package P from other nodes and is not the destination of P, ID_i will forward this package to its destination directly if the destination of P is one of its one-hop neighbors. Otherwise, if ID_i is a CH, go to step 2. If ID_i is an intercluster forwarding node, go to step 3. If ID_i is a common cluster member node, go to step 4.

Step2. If ID_i is a CH, it will find a node ID_k that is closest to the destination of P among the forwarding nodes within its cluster and its neighboring cluster heads and forward package P to ID_k.

Step3. If ID_i is an intercluster forwarding node, it will select a node ID_q that is closest to the destination of P among its CH and neighbors belonging to other clusters. Then, it forwards P to ID_q.

Step4. ID_i forwards P to its CH directly.

IV. Experiment and Discussion

In this section, we will show the simulations and analysis that were undertaken to verify the validity of the proposed framework. All of the simulations were conducted on a desktop with an Intel(R) Core(TM) i5-4590 processor, with a CPU frequency of 3.30 GHz and 4 GB memory. We implemented the proposed scheme in the discrete event-based simulator OPNET. During the simulation, a reference point group model is used to define network behaviors under mobility. In addition, the Gauss Markov model is adopted as the mobility model of each node within the network, which is more in line with the real environment. To illustrate the efficiency of our scheme, we compare the proposed scheme with two state-of-the-art weighted clustering algorithms for ad hoc networks, namely, WCA [34] and DCA [35].

A. Experiment Setup

Table II lists the values of the parameters used in the simulations. In Table II, we set a larger value to E_{hf} than E_n because CHs and intercluster forwarding nodes take more forwarding tasks, which makes their energy consumption faster than other nodes.

It is obvious that the four utility factors we consider during the clustering process are related to the initial energy of each node, the density of nodes in the network and the velocity of the nodes. Therefore, we consider three different experimental scenarios in terms of the three mentioned aspects. For the first scenario considered in the simulation, the initial energy of the nodes in the network follows a uniform distribution. Each node is evenly distributed in the simulation area. In addition, the speed of each node is the same. In the second scenario, we set the initial energy and speed of all nodes to be the same and the node density in the simulation area to be different. For the third scenario, all of the nodes have the same initial energy and different speeds, and all of the nodes are evenly distributed in the simulation area. The characteristics of the three scenarios considered in our experiments are briefly listed in Table III.

TABLE II
VALUE OF SIMULATION PARAMETERS

parameter (unit)	default value
N	10, 20, 50, 100
simulation area size(km^2)	10×10
simulation time(s)	1500
velocity of nodes(m/s)	[30,50]
communication radius(km)	1
data transmission rate(kbps)	1000
feedback collection cycle(s)	2
E_i^0(J)	2000
E_{hf}(J/s)	2.0
E_n(J/s)	1.0
α	0.8
γ	0
threshold σ	0.1
thresholdφ	2

TABLE III
SIMULATION SCENARIOS

Scenario	Description
case 1	The initial energy of each node in the network is different.
case 2	The node density in the simulation area is different.
case 3	The speed of each node in the network is different.

B. Performance Measures

We validated the performance of our scheme in terms of the clustering efficiency, the topology stability, the energy efficiency and the quality of service. Each of them is measured by the performance measures. In the following, we will explain these measures in detail.

1. Time consumption for cluster creation is used to measure the clustering efficiency. It refers to the time from network initialization to the completion of the first clustering process.
2. We use the average number of role changes per node to indicate how frequently clusters restructure. It is obvious that the smaller the number of clusters that restructure, the more stable the topology of the network is.
3. To measure the energy efficiency, we record and analyze the energy consumption of each node of the network and the lifetime of the whole network.
4. For the quality of service, we adopt the number of isolation clusters (clusters with no more than two members), the end-to-end delay of packages and the package delivery ratio as indicators.

C. Experiment Results and Discussion

In this section, we will give the simulation results for the measures mentioned above and discuss the performance of our scheme compared with the other two opponents.

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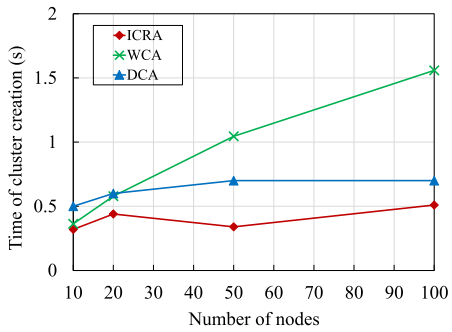

Fig. 3. Time consumption for cluster creation with $N = 100$ under case 1.

1) Clustering Efficiency: Fig. 3 shows the time consumption of cluster creation in case 1. It can be seen that our ICRA consumes less than 0.5 s to complete a clustering process. The time needed for the DCA scheme to complete a clustering is slightly more than our ICRA because in the DCA scheme, there are six factors that must be calculated in the clustering process, which will increase the time consumption. In addition, the clustering efficiency of both our ICRA and CDA schemes is not influenced by the number of nodes in the network. The WCA scheme takes the longest time to complete a clustering process. The more nodes there are in the network, the more time needed to complete clustering. The reason is that in the WCA scheme, it could require more than one CH election process to complete a clustering process, and the number of CH election processes that is required is increased with the growth of the network scale.

2) Topology Stability: Fig. 4 presents the average number of role changes of the nodes in different scenarios. In each scenario, the average number of role changes of the nodes is between 10 and 20. It is obvious that the performance of our ICRA remains stable in all three scenarios regardless of the number of nodes in the network. For the DCA scheme, the least number of role changes is more than 50, and the average number of role changes of the nodes increases slightly with the increase in the number of nodes in the network. In the three scenarios considered, the least number of role changes is more than 150 when the number of nodes in the network is no more than 100. In addition, the average number of role changes of network nodes in the WCA scheme is inversely proportional to the number of network nodes.

Given the number of nodes in the network, the average number of role changes of the nodes in our ICRA scheme is almost the same in all three considered scenarios. For the DCA and WCA schemes, the average number of role changes of the nodes varies greatly in different scenarios. For example, when there are 100 nodes in the network, the average number of role changes of the nodes in the DCA scheme is approximately 75 and 100 in cases 1 and 2, respectively, and that in the WCA scheme is approximately 170 and 250, respectively.

3) Energy Efficiency: To show the energy efficiency of our ICRA, we measure the energy consumption of each node in the network. We assume that there are 100 nodes in the network and that the initial energy values of each node are evenly distributed between 500 and 2000. Fig. 5 gives the residual energy distribution of each node at different time slots in case 1. Fig. 5a, 5b and 5c are the results of our ICRA, WCA scheme and DCA scheme, respectively. At 1000 seconds, the number of nodes with residual energy greater than 500 is 28, 25 and 7 for ICRA, WCA and DCA, and the number of dead nodes is 35, 44 and 39, respectively. At 1500 seconds, it can be seen from Fig. 5c that all nodes are dead in the DCA scheme. In our ICRA, the number of nodes whose residual energy is more than 400 is 14, which is greater than that of

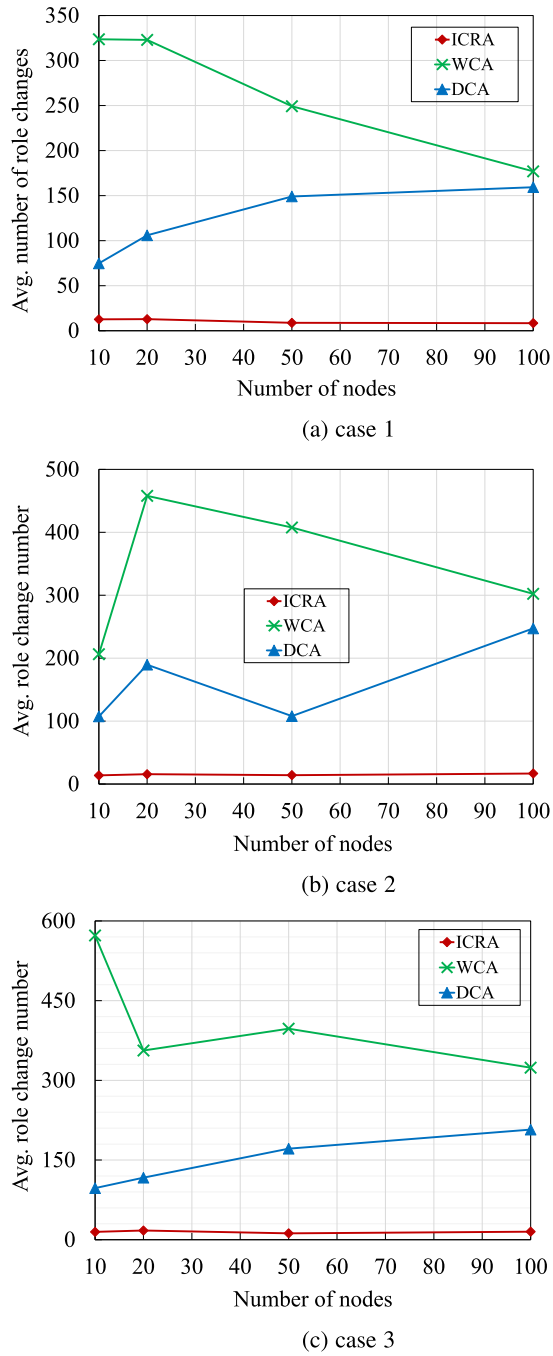

Fig. 4. Average number of role changes of the nodes in the network in different scenarios.

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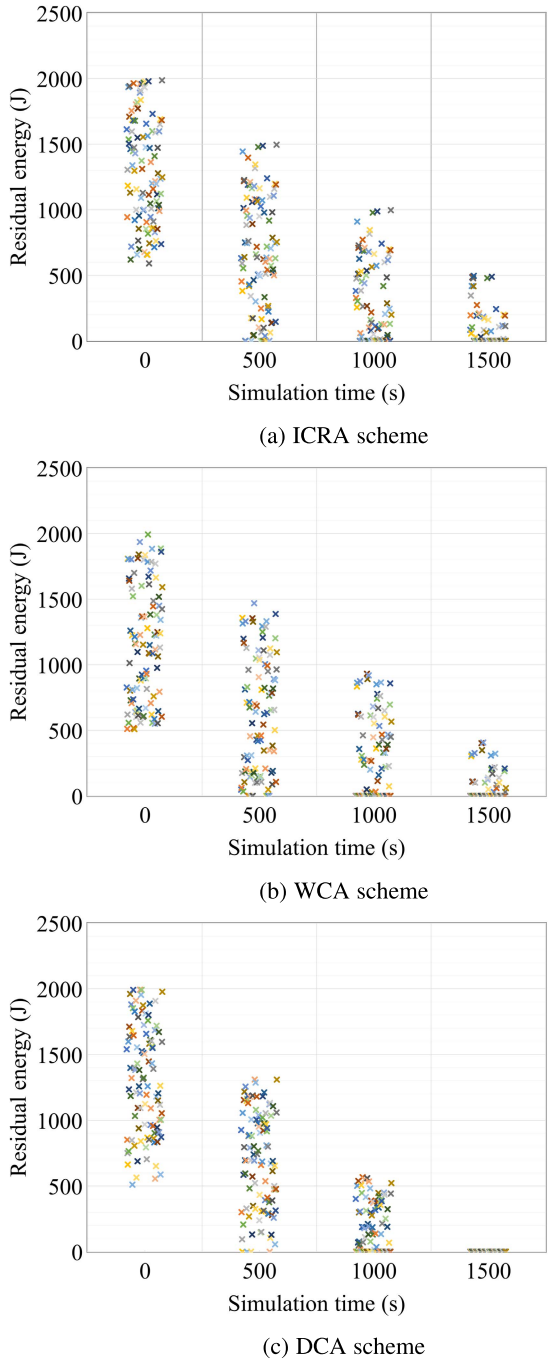

Fig. 5. Residual energy distribution of the nodes in case 1 with $N = 100$.

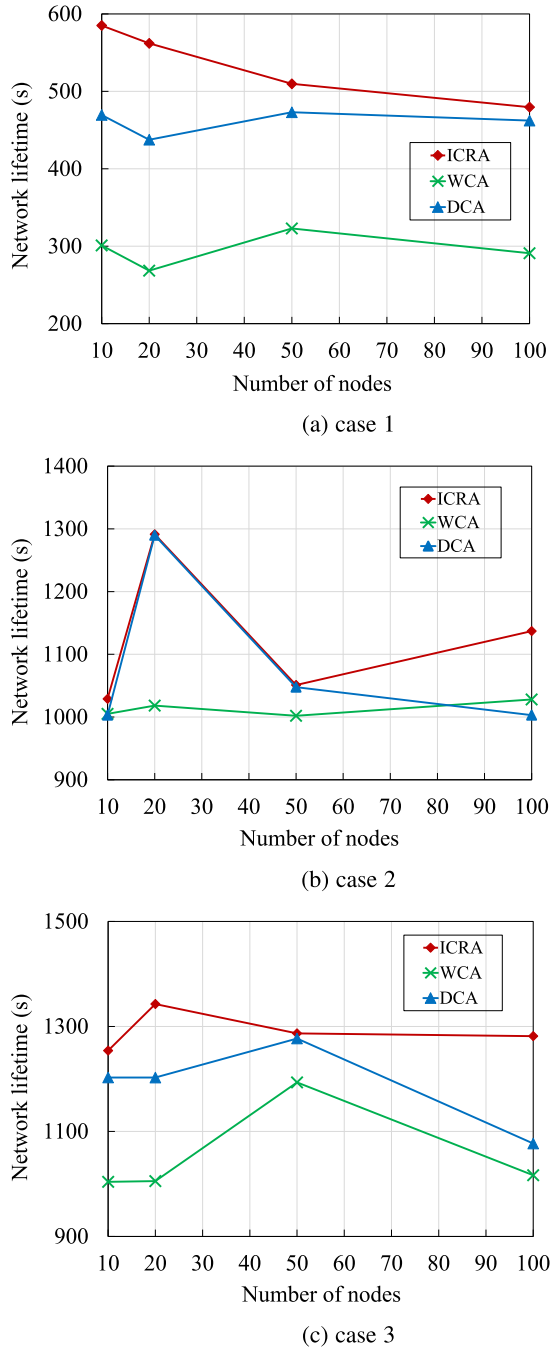

Fig. 6. Comparison of the network lifetime in different scenarios.

the WCA (13). In addition, the number of dead nodes in our ICRA (68) is less than that of the WCA scheme (72).

In Fig. 6, we compare the death time of the first dead node (we call it the network lifetime here) in the network to illustrate the energy efficiency of each scheme. From Fig. 6a, it can be seen that in case 1, the network lifetime of ICRA is longer than that of the other compared schemes.

In cases 2 and 3 (shown in Fig. 6b and 6c), ICRA is still the scheme with the latest death time of the first dead node. Overall, in cases 2 and 3, the network survives longer. The reason is that in both cases, the initial energy of each node is 2000, while in case 1, the minimum initial energy of the nodes in the network is 500.

4) Quality of Service: First, we analyze the quality of service in terms of the node density in the network. Fig. 7 shows the number of isolation clusters for the compared three schemes in case 2. Here, we call the cluster with no more than two members the isolation cluster. The more isolation clusters in the network, the longer the communication delay will be. Because the probability of intercluster communication between nodes will become larger, we know that intercluster routing requires more hops to be completed.

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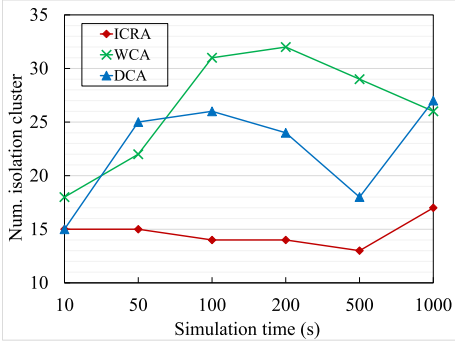

Fig. 7. Comparison of the number of isolation cluster in case 2.

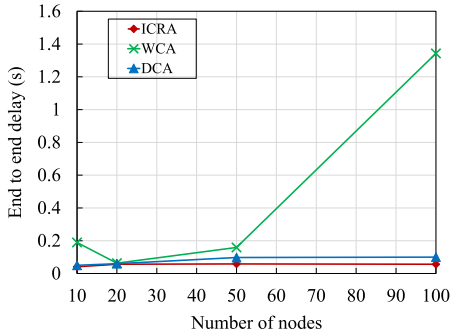

Fig. 8. Comparison of the end to end delay of packages in case 1.

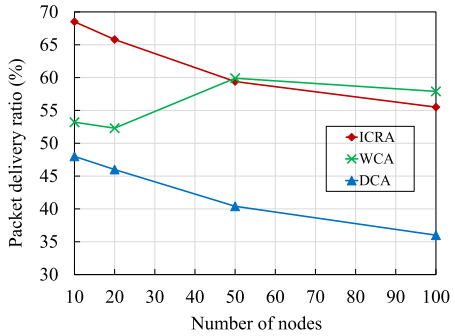

Fig. 9. Comparison of the packages delivery ratio in case 3.

It can be seen from Fig. 7 that when there are 100 nodes in the network, the number of isolation clusters for ICRA is stable at approximately 15 during the whole simulation period. The WCA and DCA schemes show $3 \sim 18$ and $0 \sim 12$ more isolation clusters than ICRA. The reason is that ICRA can adjust the clustering strategy under different network states.

Fig. 8 gives the results for the end-to-end delay of packages in case 1. The performance of ICRA is similar to that of DCA, which is not influenced by the number of nodes. For the WCA scheme, when the number of nodes reaches 100, the end-to-end delay of packages will rise sharply. The reason is that the WCA scheme allows only the CH to forward packages to other clusters. There are no intercluster forwarding nodes that can forward packages among its cluster and its neighboring clusters, and thus, intercluster communication could take more time.

Because the link stability between nodes is highly correlated with their speeds, we measure the performance of the compared three schemes in terms of the package delivery ratio in case 3. From Fig. 9, it can be seen that the packet delivery ratio decreases slightly with an increasing number of nodes in the ICRA and DCA schemes. The performance of ICRA and WCA is similar when the number of nodes is more than 50. The DCA scheme has the lowest packet delivery ratio because it considers only the instant link quality during the clustering process. In the scenario in which the speed of the nodes varies constantly, the link quality is changing all the time, and thus, the available link could invalid when sending packets.

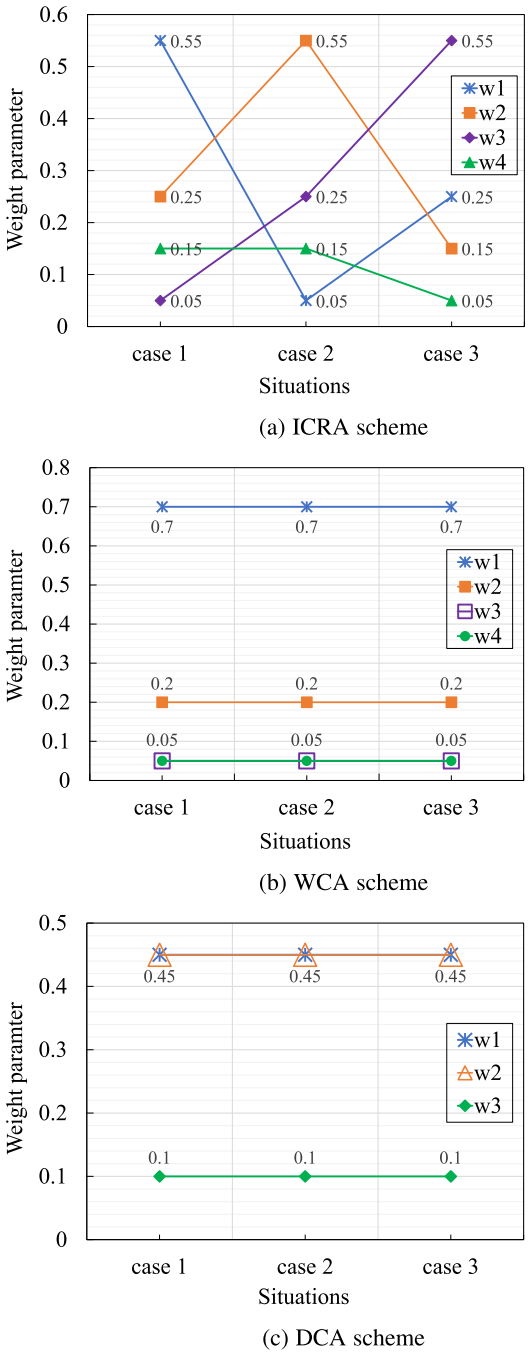

Fig. 10. Comparison of the weight evolution for utility factors.

Fig. 10 shows the evolution of the weighting parameter for each factor considered in the utility calculation process. From Fig. 10a, it can be seen that the weight of each utility factor varies in different scenarios, which is the reason for the outperformed performance of our scheme. For the other

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two schemes, the value of the weighting parameters will not change with the changing scenarios. Here, we adopt the value of the weight parameters adopted by the authors of the two schemes used in their experiments. For the WCA scheme shown in Fig. 10b, w1, w2, w3, and w4 represent the number of neighbors of a node, the degree difference of a node, the sum of the distance of a node with its neighbors, and the running average speed of a node, respectively. Parameters w1, w2, and w3 in Fig. 10c mean the energy, LQIR (link quality improvement ratio) and SCDR (self-to-common degree ratio), which the authors adopted in the experiment in the mobile scenario for stability goal.

V. Conclusion

In this paper, we propose an intelligent routing protocol for UANETs using reinforcement learning model. The whole protocol is composed of a clustering process and routing process. During the clustering process, we measure the utility of each node to select the CH, which is the weighted sum of four utility factors. By using the reinforcement learning model, the weights of the four utility factors can be adjusted with the changing network state, so that to maintain a stable network topology. For the routing process, we give the intra-cluster routing and inter-cluster routing method based on the network topology. In order to verify the performance of our method, we compare our method with other two available schemes in terms of their clustering efficiency, topology stability, energy efficiency and quality of service. The results show that our scheme takes less time to complete the clustering process. For other measures, our scheme also achieves better performance compared to its state-of-the-art counterparts.

In the future, we will consider the security of the messages transmitted during the clustering process. Because the most available works about the routing security focus on the security of the routing process, while if the clustering packets are tampered by adversaries, the network topology may suffer severe frontal deviation from that it should be, which will lead to unstable and inefficient routing process.

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